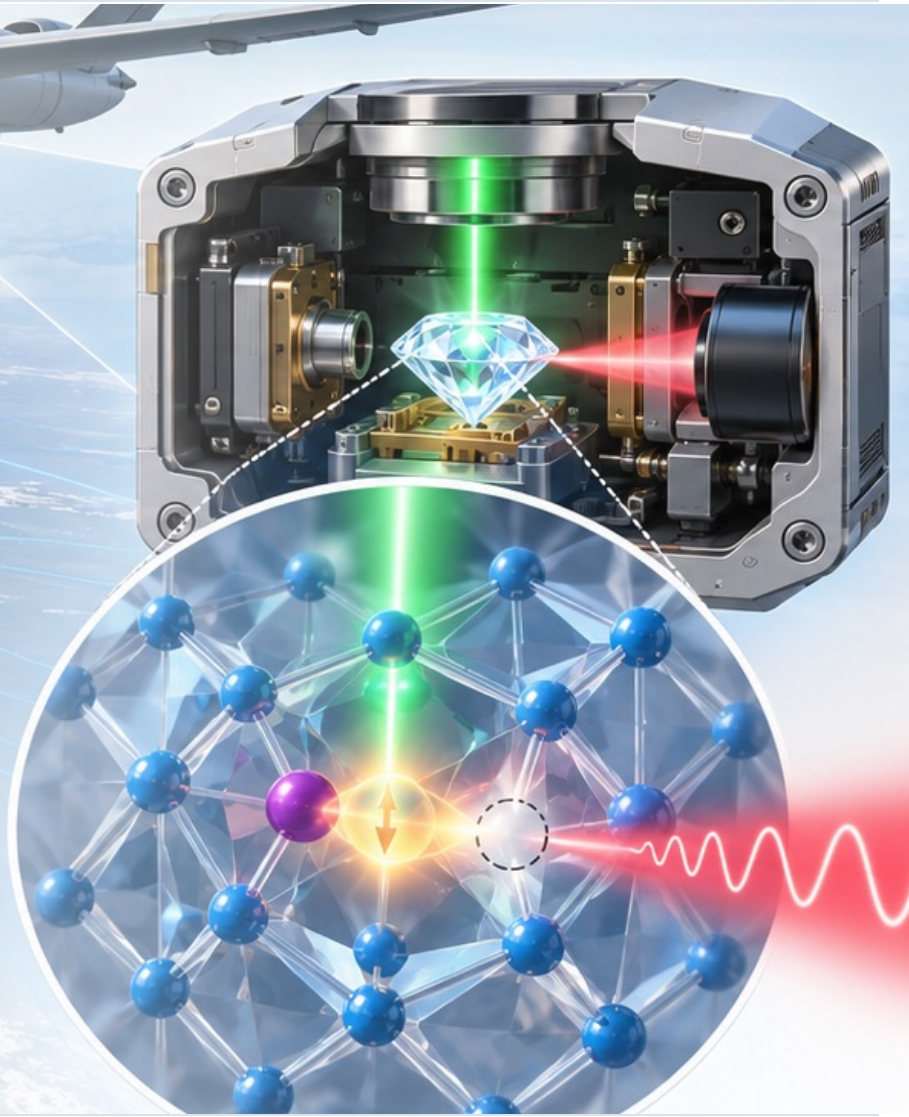
Entanglement-Dynamic Topology for Robust Quantum Sensing



Finding the finite control window where sensitivity, coherence, fidelity and readout remain jointly effective.

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National Research Council Canada · Defence Research and Development Canada

Introduction: defence applications

Magnetic and low-frequency RF signatures resist masking and reach where optical sensing fails — exploiting them needs lab sensitivity in a fielded package.

Magnetic anomaly detection

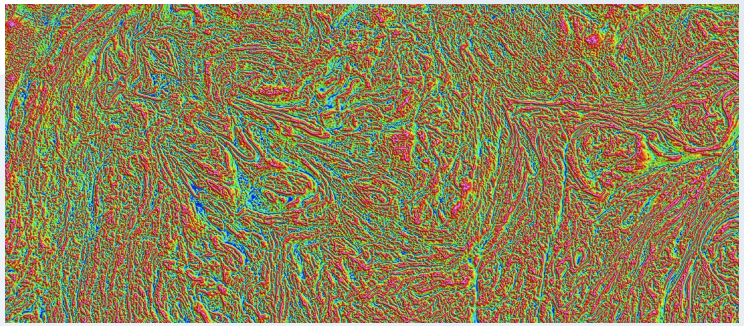
A submerged steel hull distorts Earth's magnetic field. MAD instruments detect the local anomaly.

GPS-denied Magnetic navigation

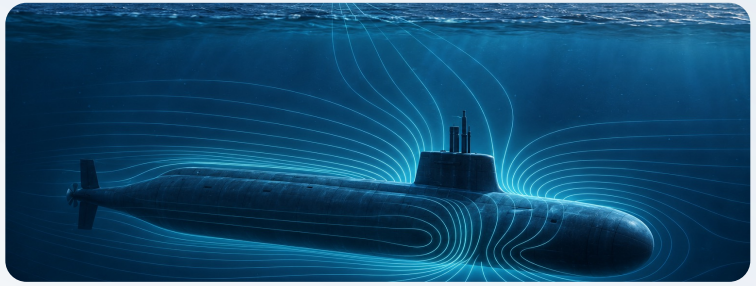
Anomaly maps give an unjammable position fix where satellite navigation is denied.

The hard part is not peak laboratory sensitivity — it is holding a known operating point in the field.

SBQuantum magnetic map



Source: SBQuantum, Magnetic Navigation, sbquantum.com



Introduction: background information

A nitrogen-vacancy defect in diamond is an optically addressable electron spin: initialise with green light, evolve in the field, read out the fluorescence.

$S = 1$

Spin triplet

Zeeman splitting shifts the m = ±1 levels linearly with the field along the NV axis.

$\varphi \propto B \cdot T$

Phase, not amplitude

Sensitivity improves with interrogation time, so coherence is the resource that sets it.

$T_2^* \approx \mu s$

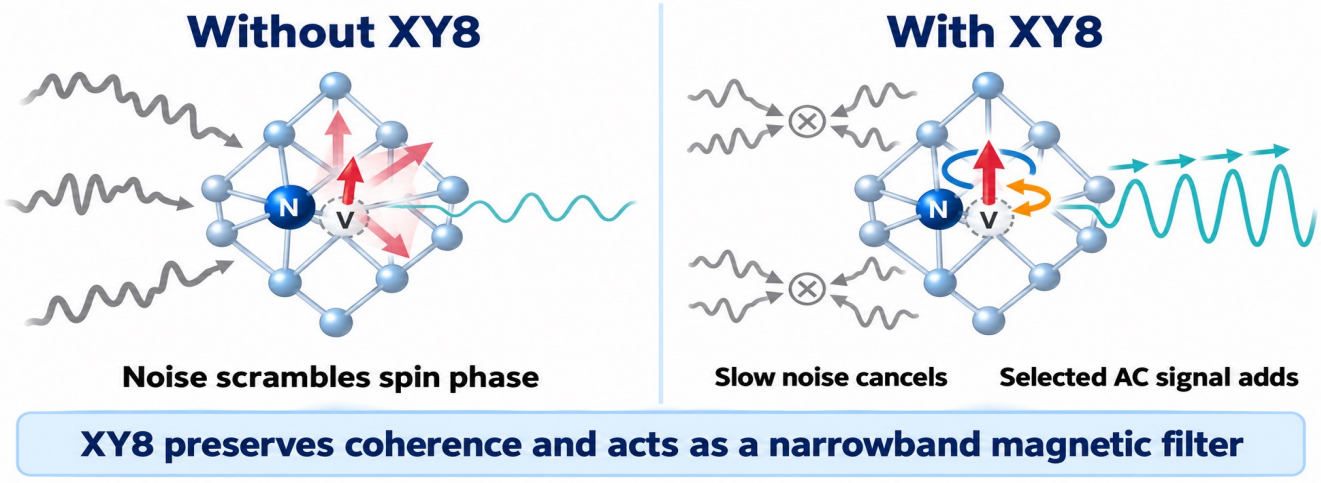
The limit to beat

Bath-induced dephasing ends free evolution in microseconds, far short of the phase budget.

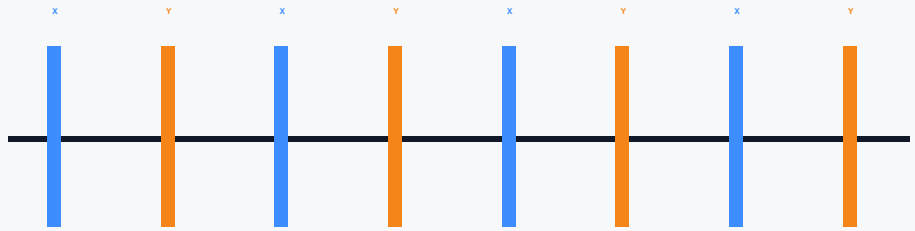
Every sensitivity gain here comes from extending usable interrogation time without losing the signal.

Introduction: background information

A periodic train of π pulses refocuses slow bath noise and makes the sensor respond only near a chosen frequency — extending coherence and adding selectivity at once.



One XY8 block



T pulse spacing sets the sensing band

N blocks → 8N π pulses

At resonance: $\tau = 1 / (4f)$

Introduction: background information

In weak-field magnetometry and narrowband RF sensing, useful control resources compete with the costs of running them.

↑ signal

Phase accumulation

Deeper XY8 sequences accumulate more target phase and sharpen the filter.

↑ selectivity

Noise rejection

Dynamical decoupling suppresses environmental noise around the sensing band.

↑ cost

Errors + time

Pulse infidelity, decoherence exposure and overhead accumulate with depth.

Design objective: minimize sensitivity penalty η(τ,N) under all resource constraints—not maximize N.

What EDT Means in This Study

EDT maps where sensing performance and the required quantum resources remain jointly useful.

ENTANGLEMENT

Broader framework

Multi-spin entanglement may improve field response, but it also adds preparation and coherence costs.

Current study: one NV spin
Future work: coupled or entangled spins

DYNAMICS

Calculated here

XY8 reshapes the sensor response during a measurement.

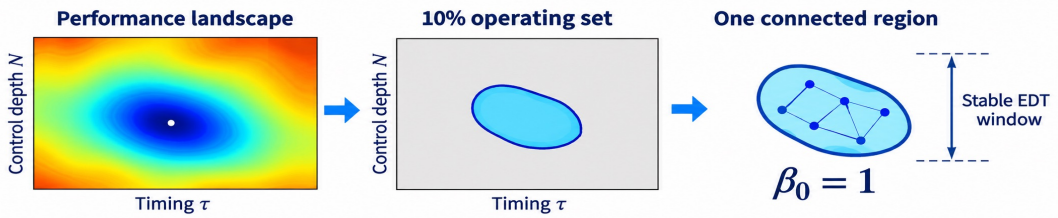
τ selects the sensing frequency
N controls accumulation, selectivity, errors and duration

TOPOLOGY

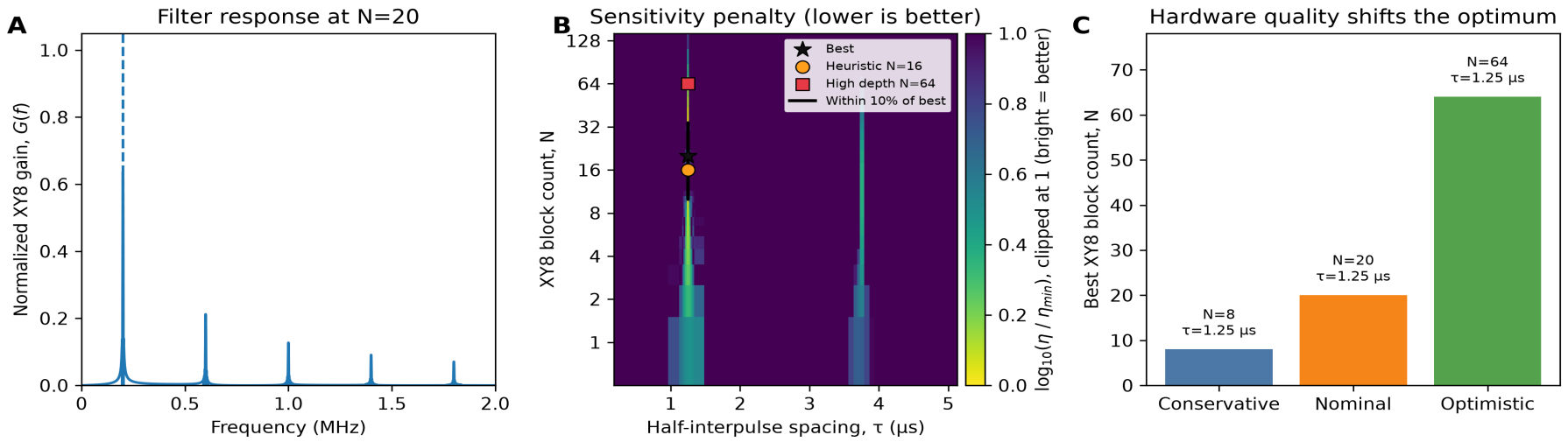
Calculated here

Near-optimal set:
$\eta(\tau,N) \leq 1.1\eta_{min}$

At 200 kHz: one connected resonant region, $\beta_0=1$, stable under timing-grid refinement



Refined analysis resolves a finite, hardware-dependent control window



NOMINAL OPTIMUM

N=20, τ=1.25 μs

160 microwave pulses

10% OPERATING WINDOW

N=10–34

Within 10% of the minimum

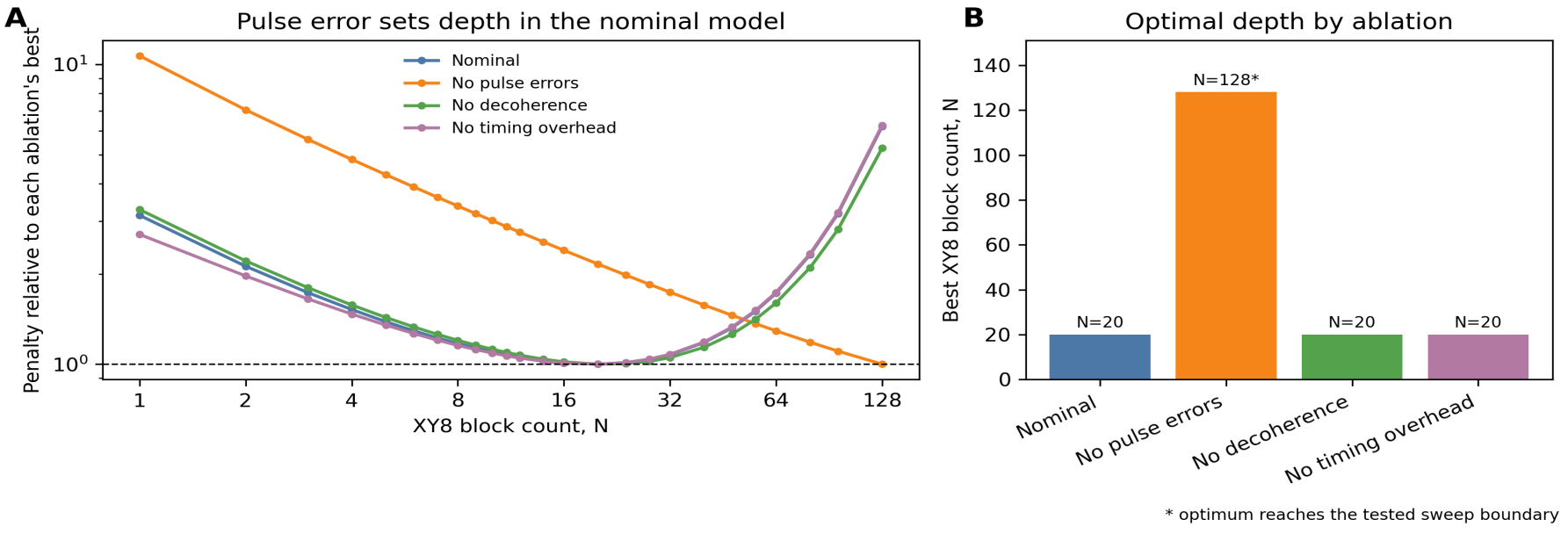
HARDWARE DEPENDENCE

Optimal N: 8–64

Conservative to optimistic hardware

The early coarse scan suggested N=4–10. Exact resonance sampling and a denser depth sweep resolve N=20 and a broader N=10–34 near-optimal window.

Pulse Errors Create a Finite Optimal Control Depth



MECHANISM

Remove pulse errors

20 → 128*

~71% lower η

Remove decoherence

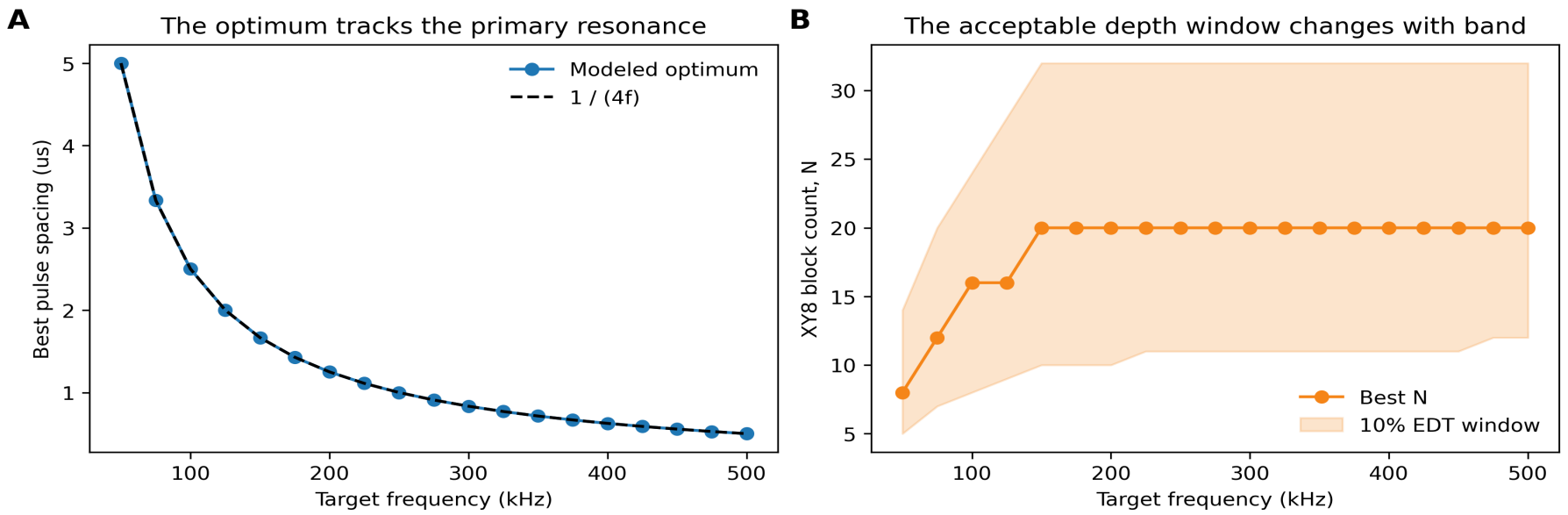
N stays 20; η improves ~4.8%

Remove overhead

N stays 20; <1% change

Pulse-error accumulation creates the interior optimum

Results: Pulse Timing Selects the Target Sensing Frequency



Timing tracks the sensing band

DIRECT RULE

$$\tau = 1 / (4f)$$

50–500 kHz

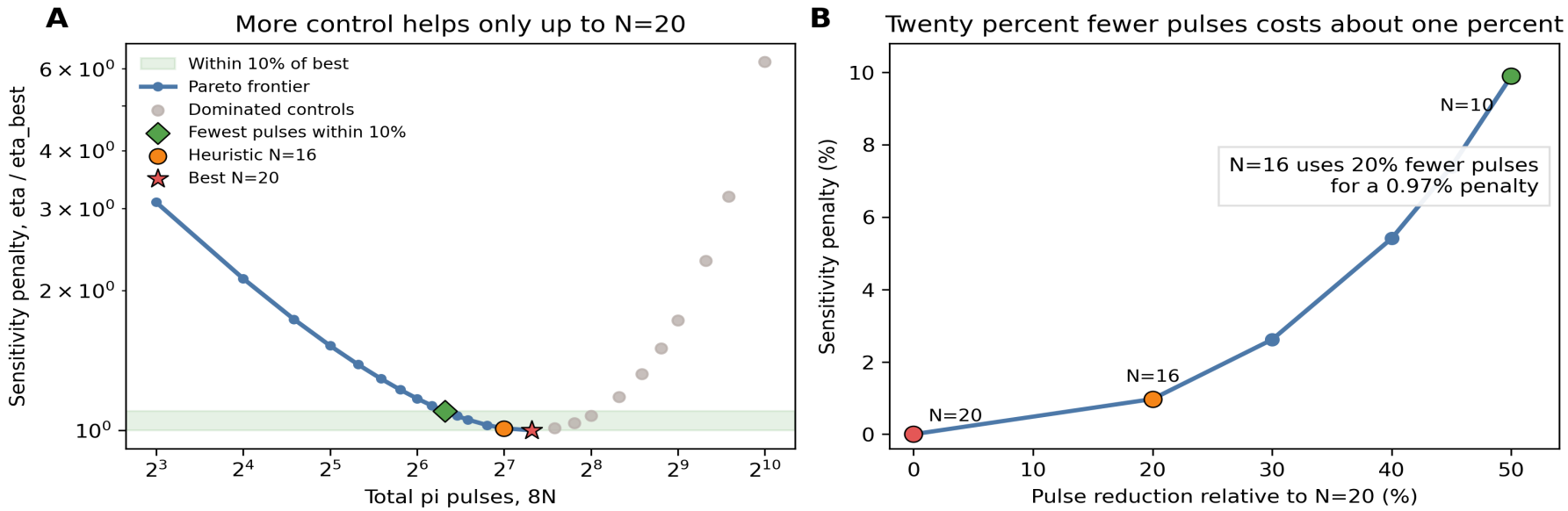
Modeled τ follows the analytic primary resonance.

Best depth

N = 8 → 20

The 10% window changes with band.

N = 16 Preserves Performance with Fewer Pulses



PRACTICAL CHOICE

N = 16

128 π pulses

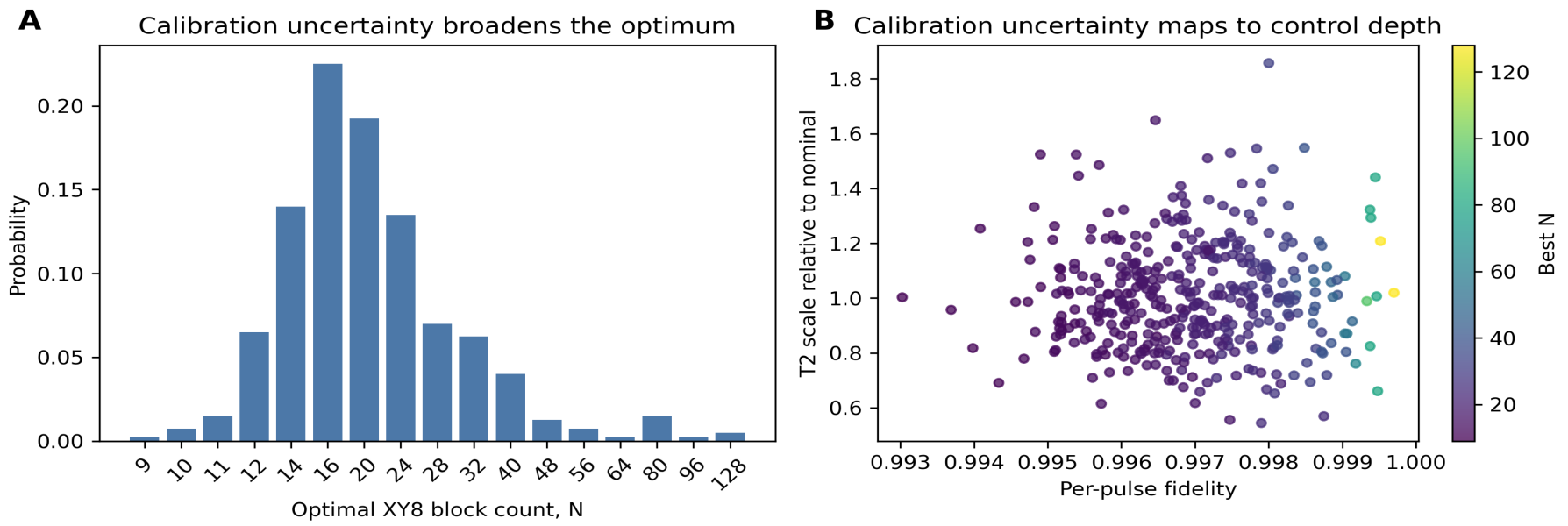
−20%

pulse count

+0.97%

sensitivity penalty

Hardware Uncertainty Broadens the Preferred Control Depth



Uncertainty broadens the preferred depth

400

calibration samples

89%

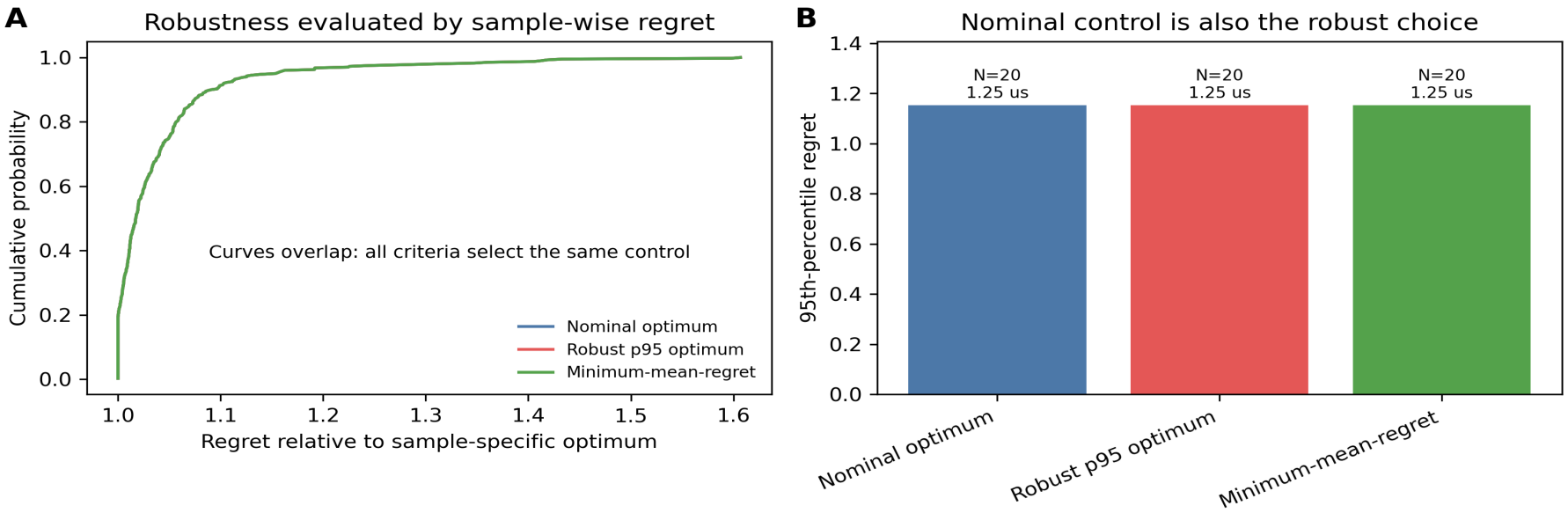
select N = 12–32

Mode

N=16

22.5% of samples

N = 20 Is the Most Robust Fixed Choice



ALL THREE CRITERIA

N = 20

τ = 1.25 µs

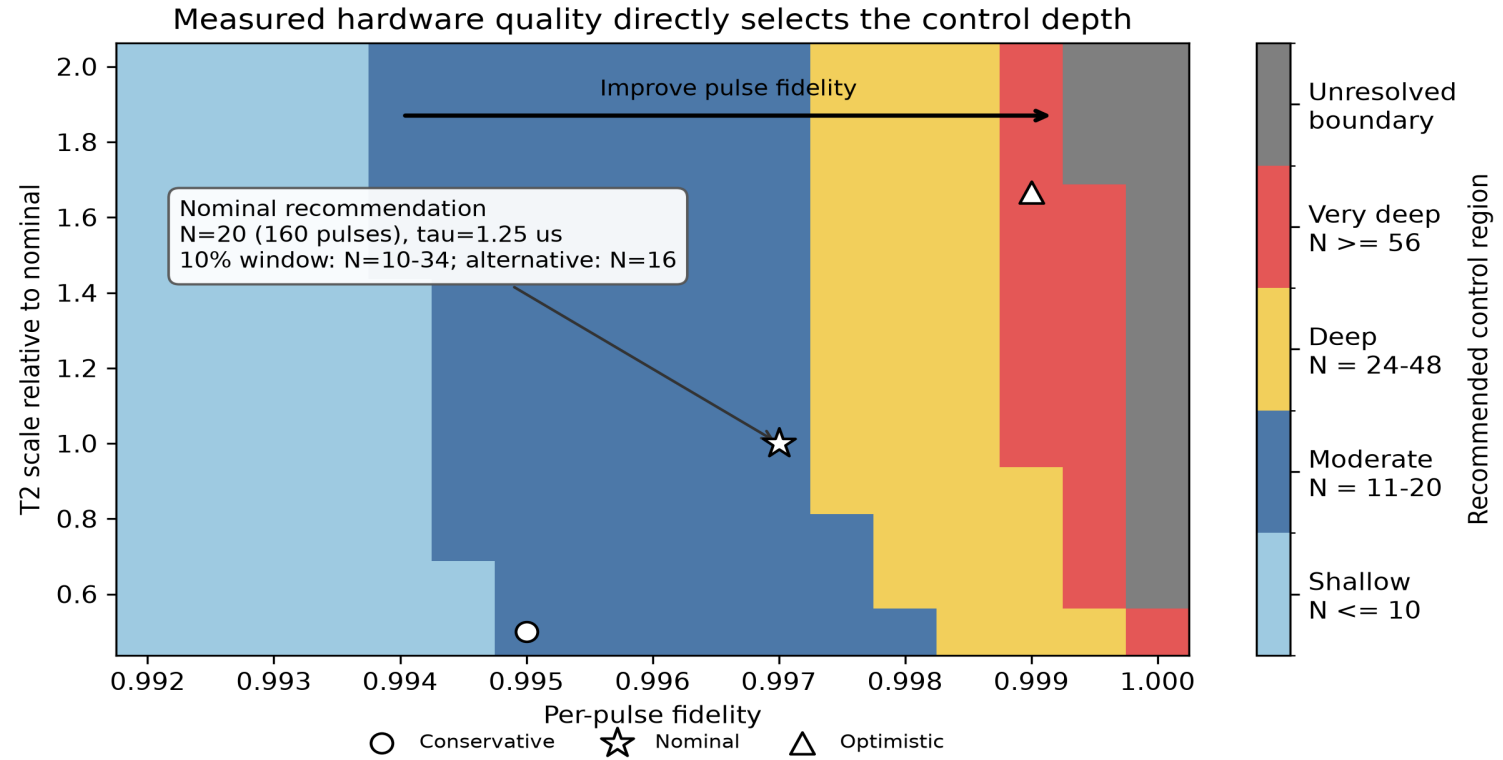
Nominal

95th-percentile regret

Minimum mean regret

p95 regret = 1.154×

Measured Hardware Quality Selects the Control Depth



Scenario markers project only F_pi and T2; contrast, overhead, photon rate, and coherence scaling remain nominal in the map.

DECISION RULE

$$(f, F\pi, T_2)$$

↓

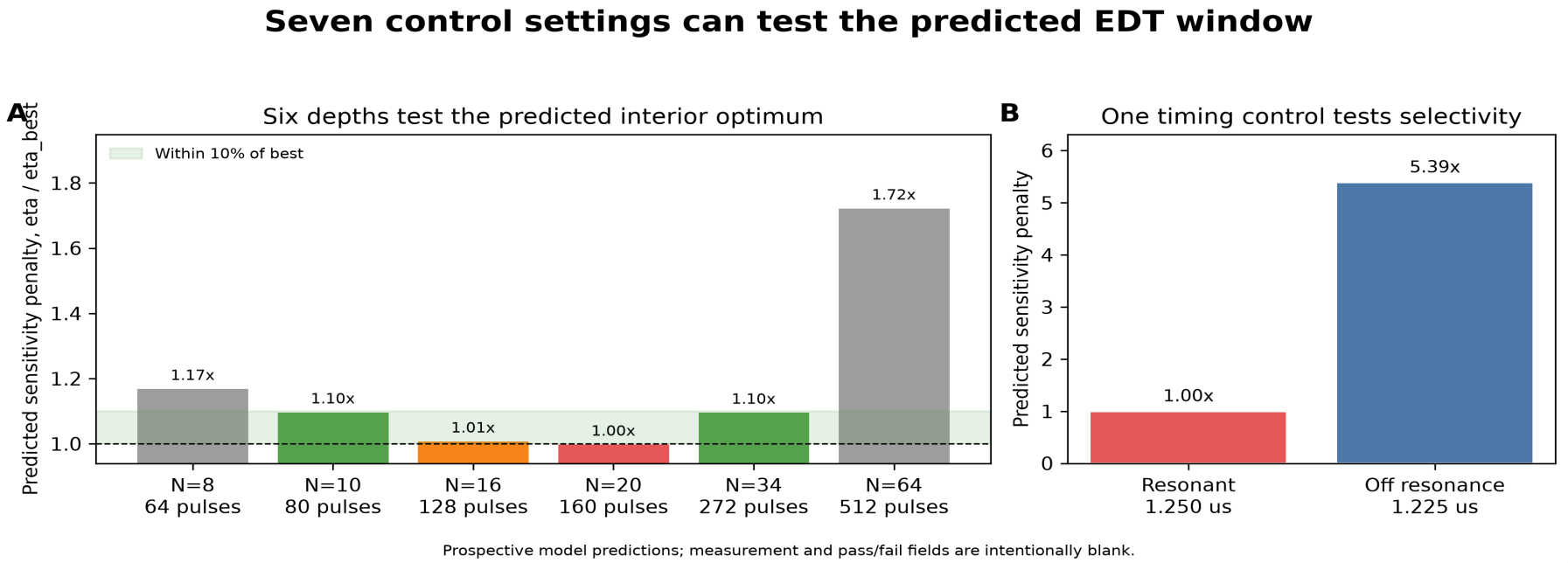
(τ, N, 10% window)

Nominal hardware

τ = 1.25 µs
N = 20 · 160 π pulses
window N = 10–34
Pareto option N = 16

Caveat: contrast, photon rate, overhead and other coherence parameters are held nominal.

Seven measurements can test the predicted EDT window



FALSIFIABLE TESTS

1. N=16 and N=20 are tied or ordered as predicted
2. N=8 and N=64 are worse than interior controls
3. Measured optimum lies within N=10–34
4. Off-resonance is worse than resonance
5. Values fall within model intervals

Prediction table generated before measurements are entered.

First-principles data connect material physics to control

The diamond host is fixed, but temperature and strain change the NV transition and relaxation rates used by the control model.

Spin–strain

Udvarhelyi et al.

Six DFT coefficients map axial, transverse and shear strain into the ground-state spin Hamiltonian.

Transition shift

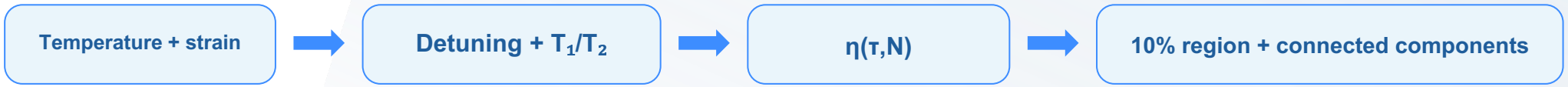
Cambria + Tang

The analytical D(T) law supplies detuning. Tang's first-principles slope, −75.99 kHz/K, provides an independent cross-check.

Spin–phonon relaxation

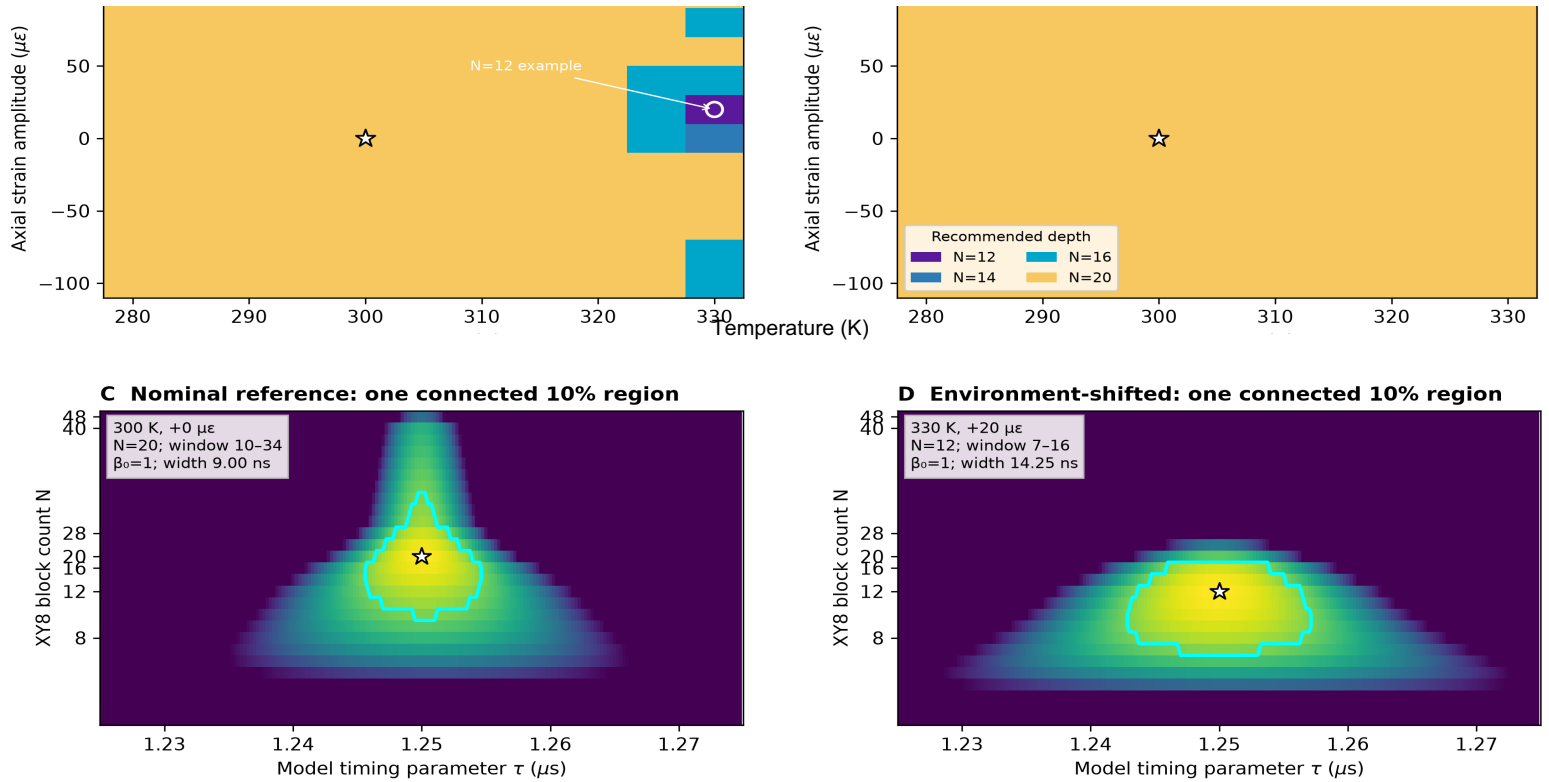
Mondal et al.

The ab initio $T_1(T)$ curve constrains the temperature-dependent coherence correction without fitting the EDT optimum.



Inputs were frozen before the environment scan. No parameter was tuned to the EDT result.

Environmental detuning moves the optimum without splitting the EDT region



CONFIRMED RESULT

Fixed carrier

N = 12

330 K, +20 με example

Carrier retracked

N = 20

throughout the displayed slice

Topology

$\beta_0 = 1$

226 / 226 fine-grid confirmations passed

Environment changes the region's geometry and recommended depth while its connected-region count stays fixed

EDT makes control depth a testable decision

N=20

Resolved optimum

At 200 kHz: τ=1.25 μs and a 10% window spanning N=10–34.

Pulse error

Dominant depth limiter

Removing pulse errors pushes the optimum to the tested boundary.

7 settings

Prospective validation

A minimal experiment can confirm or falsify the predicted window.

Next step: measure the seven-point protocol, update the hardware calibration, then let the decision rule select control depth.